ADEBAYO A. OMOLUMO

adebayoomolumo@gmail.com
LinkedIn: linkedin.com/in/adebayo-omolumo | GitHub: github.com/Bayurzx | Portfolio: work.adebayoomolumo.website

SUMMARY

Highly accomplished and results-oriented Senior DevOps Engineer and Software Developer with 8+ years of experience in designing, implementing, and managing robust cloud infrastructure, CI/CD pipelines, and scalable software solutions. Proven expertise across AWS, Azure, and GCP environments, specializing in Infrastructure as Code (IaC), automation, containerization (Docker, Kubernetes), and serverless architectures. A certified Project Management Professional (PMP) and multi-cloud certified expert (AWS, Azure, GCP) with a strong track record of leading successful projects, optimizing system performance, enhancing security, and driving operational efficiency. Passionate about leveraging cutting-edge technologies to solve complex engineering challenges and foster innovation.

SKILLS

- **Cloud Platforms:** AWS (EC2, S3, Lambda, CloudFormation, RDS, EKS, CloudFront, IAM, DynamoDB), Microsoft Azure (VMs, App Service, Azure Functions, AKS, Cosmos DB, Azure DevOps, Sentinel, Logic Apps, Event Hubs), Google Cloud Platform (GCP) (Compute Engine, GKE, Cloud Storage, Cloud Functions, BigQuery, Vertex AI, Cloud Run)
- **DevOps & CI/CD:** Jenkins, CircleCI, Azure DevOps, GitHub Actions, Docker, Kubernetes, Ansible, Terraform, CloudFormation, Prometheus, Grafana, ELK Stack, Git
- **Programming & Scripting:** Python (Flask, Django), JavaScript (React.js, Next.js, Vue.js, Node.js), TypeScript, Solidity, Bash, PowerShell, HTML5, CSS3
- **Databases:** PostgreSQL, MySQL, MongoDB, DynamoDB, Cosmos DB, SQL Server
- **Operating Systems & Networking:** Linux (Ubuntu, CentOS), Windows Server, TCP/IP, DNS, VPN, Load Balancing, Firewalls
- **Security & Compliance:** IAM, Network Security, Data Encryption, SIEM (Azure Sentinel), Security Best Practices, PCI-DSS Awareness, Vulnerability Management, Zero Trust Principles
- **Tools & Methodologies:** Agile, Scrum, Jira, Confluence, Microsoft Office Suite, Project Management

PROFESSIONAL EXPERIENCE

Cloud DevOps Engineer | INFOMETICS | Remote
September 2024 – Present

- **Architected and deployed a production-grade, highly available Kubernetes testbed on AWS, simulating an air-gapped, on-premise environment for stateful IBM workloads (ACE/MQ).**
    - **Infrastructure as Code (IaC):** Provisioned a multi-node (1 master, 3 worker) Kubernetes cluster using Terraform, featuring automated node bootstrapping with kubeadm and secure, private networking via a bastion host topology.
    - **Distributed Storage Engineering:** Implemented a resilient, software-defined storage layer using Longhorn, providing replicated block storage for all stateful applications. Systematically diagnosed and resolved storage scheduling failures by right-sizing node disk capacity based on total replicated volume requirements.
    - **Stateful Workload Orchestration:** Re-architected IBM ACE and MQ applications from simple containers to robust StatefulSets, ensuring stable network identity and leveraging Longhorn for persistent, highly available data storage.
    - **Secure & Observable Networking:** Engineered a comprehensive ingress and pod-to-pod communication strategy using NGINX Ingress and Calico CNI. Implemented a zero-trust network model by authoring granular NetworkPolicies to secure application traffic.
    - **Disaster Recovery & Backup:** Designed and implemented a complete disaster recovery solution using Velero with Kopia, backing up both Kubernetes manifests and persistent volume data to an in-cluster Minio S3 store. Authored and validated an automated DR test script to simulate and recover from a full namespace deletion.
    - **Full-Stack Observability:** Deployed and configured a comprehensive monitoring stack with Prometheus for metrics collection and Grafana for visualization, enabling real-time insights into cluster and application health, including custom dashboards for IBM MQ queue depth and throughput.
    - **Automated Certificate Management:** Hardened cluster security by implementing automated TLS certificate lifecycle management, using kubelet-csr-approver for node bootstrap and cert-manager for application-level certificates.

- Lead the setup and management of testing and development environments, ensuring scalable cloud infrastructure, seamless workflows, and optimized resource allocation with security best practices.
- Develop automated backup systems for code repositories (e.g., using AWS Lambda and Docker), ensuring weekly backups with robust logging and security compliance.
- Migrate frontend, backend, and database Docker applications to secure networks, maintaining compliance with relevant security policies.

- Implement automation on cloud platforms (e.g., AWS) to manage testing databases and instances, significantly reducing resource usage and associated costs.
- **Key Client Engagement: UBA Group (United Bank for Africa), Lagos**
    - Engineered and deployed containerized (Docker) test environments for Nginx, frontend, and backend applications within UBA's secure, isolated internal network.
    - Configured and managed server installations with all necessary dependencies, enabling multiple distinct projects to run concurrently on single test servers via containerization, each on dedicated ports.
    - Spearheaded the seamless migration of applications from the meticulously prepared test environment to the live production environment.
    - Led the critical upgrade of the project's network infrastructure to HTTPS, significantly enhancing data security and integrity.
    - Designed and implemented a custom deployment pipeline to build, package, and securely transfer frontend and backend artifacts to the air-gapped bank environment, overcoming internet access restrictions.
    - Collaborated extensively with UBA's security team to implement and enforce stringent security best practices, including HSTS, SSL certificate management, secure http-only cookies, and comprehensive Nginx security policies.

Founder & Lead Cloud/DevOps Engineer | IGLUM TECH SERVICES (formerly Iglum Nig. Ltd.) | Ikorodu, Lagos
September 2019 – August 2024

- Founded and led a tech services agency, evolving from full-stack web development to specialized cloud engineering and DevOps consulting for various clients.
- Led the design and deployment of cloud infrastructure on AWS, Azure, and GCP, ensuring scalability, security, and compliance with industry standards for client projects.
- Architected and implemented CI/CD pipelines using Jenkins, CircleCI, and Azure DevOps for automated build, test, and deployment of web applications and microservices, reducing deployment time by up to 50%.
- Managed migration of on-premise infrastructure to Azure for clients, optimizing resource allocation and achieving average annual cloud expense reductions of 30%.
- Developed and launched "Save Earth," an AI-powered platform for endangered species awareness, integrating Azure Custom Vision and Blob Storage; secured two hackathon wins and AWS funding.

- Redesigned and migrated the Iglumtech company website to AWS S3 with CloudFront CDN, implementing a CircleCI DevOps workflow for automated updates and enhanced security (HTTPS).

Computer Science Teacher | National Youth Service Corps – UEC College | Ikorodu, Lagos
September 2019 – March 2020

- Taught computer science fundamentals and programming concepts to secondary school students.

KEY PROJECTS

- **Chainlink Oracle Integration (Hackathon Winner):** Led a team to streamline Chainlink project deployments using CircleCI for CI/CD. Implemented AWS CloudFormation for infrastructure, automated rollback mechanisms, and deployed React/Next.js frontend and Solidity/Hardhat backend. (May 2022 - Aug 2022)
- **Operationalizing ML Microservice with Kubernetes (Udacity Project):** Deployed a pre-trained Scikit-learn model for housing price prediction as a microservice using Docker and Kubernetes. Integrated CircleCI for CI, improved logging, and tested with Hadolint. (Sept 2023 - Oct 2023)
- **Serverless TODO Application on AWS Lambda (Udacity Project):** Developed and deployed a serverless TODO app using AWS Lambda, Serverless Framework, DynamoDB, and API Gateway. Implemented authentication, file uploads (S3 pre-signed URLs), and adhered to best practices for code organization and monitoring (X-Ray). (Mar 2023 - Apr 2023)
- **Neighborly Microservice on Azure (Udacity Project):** Developed and deployed a Python Flask web application connecting neighbors. Utilized Azure Function App, Cosmos DB (MongoDB API), Azure Kubernetes Service (AKS), Event Hubs, and Logic Apps for a complete microservices architecture with CI/CD. (June 2021 - July 2021)
- **Azure Cloud Security & Monitoring Implementation:** Led initiatives to enhance cloud security posture for client projects using Azure Sentinel (SIEM/SOAR), Azure Security Center, IAM, network segmentation (Zero Trust), and JIT access. Implemented comprehensive monitoring using Application Insights, Log Analytics, and automated remediation via Azure Runbooks for VMSS and AKS environments. (Aug 2021 - Sept 2022)
- **Bunnyshell Hackathon (Team Lead & Winner):** Leveraged Bunnyshell's Environment-as-a-Service (Kubernetes-based) to deploy Terraform infrastructure, streamlining development, testing, and staging environments. Enabled parallel frontend/backend development and improved collaboration. (July 2023 - Aug 2023)

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT (Selected)

- **Project Management Professional (PMP®)** – PMI
- **Microsoft Certified: DevOps Engineer Expert** (Earned Aug 2023, Renewed)
- **Google Cloud Certified: Professional Cloud Security Engineer** (Earned June 2024)
- **Google Cloud Certified: Professional Cloud Architect** (Earned Nov 2022)
- **Google Cloud Certified: Associate Cloud Engineer** (Earned Apr 2022, Renewed)
- **Microsoft Certified: Azure Administrator Associate** (Earned Apr 2021, Renewed)
- **Microsoft Certified: Azure Developer Associate** (Earned Aug 2021, Renewed)
- **Microsoft Certified: Azure Fundamentals** (Earned Apr 2021)
- **Udacity Nanodegree: Cloud Developer using Microsoft Azure** (Sept 2021)
- **LFS211: Linux Networking and Administration** – The Linux Foundation (Mar 2024)
- **AWS Educate Machine Learning Foundations** (Valid through Apr 2025)
- ***Multiple Google Cloud Skill Badges*** *(Vertex AI, Security, Data Loss Prevention, etc. – 2024)*
- ***OpenHack****: DevOps – Microsoft (May 2021)*

EDUCATION

Bachelor of Engineering (B.Eng.), Wood Products Engineering
University of Ibadan | Ibadan, Oyo State, Nigeria
2015 – 2019

National Diploma, Mechanical Engineering
Lagos State Polytechnic | Ikorodu, Lagos State, Nigeria
2013 – 2015

AWARDS & RECOGNITION (Selected)

- **Bunnyshell Hackathon Winner** (Team Lead) – 2023
- **Celo's Make Crypto Mobile Hackathon Winner** (Infrastructure & Web3 Track, 1st Place) – 2021
- **Datastax Holiday Hackathon Winner** (EMEA Category) & Top 10 Finalist (4th) – 2021
- **SaveEarth Project: AWS Funding Recipient & 2x Hackathon Winner** – 2020-2022
- **Microsoft Azure AI Hackathon Top 50 Finalist** (23rd) – 2021
- **Andela Google Africa Developers Scholarship** (GCP Track Mentor) – 2020-2021
- **Bertelsmann Technology Scholarship Program** (Azure Cloud Computing) – 2020-2021